

SECTION B - LECTURE 05

Programme Evaluation and Review Technique (PERT)

Question-to-Slide Mapping for Term-Final Questions, 2016-2024

Variability in Activity Times

Estimate follows beta distribution

Expected time:

$$t_e = (a + 4m + b)/6$$

Variance of times:

$$\sigma^2 = [(b - a)/6]^2$$

Example: 10 (ref. V.K. Kapoor, pg 16.49)

• A small project consisting of eight activities has the following characteristics:

Time estimates (in weeks)

Activity	Preceding activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	4	12
B	-	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	B, C	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

Example: 11 (ref. V.K. Kapoor, pg 16.51)

• The information is given (time estimates are in days):

Activity	Preceding activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	2	2
B	-	1	3	7
C	A	4	7	8
D	A	3	5	7
E	B	2	6	9
F	B	5	9	11
G	C, D	3	6	8
H	E	2	6	9
I	C, D	3	5	8
J	G, H	1	3	4
K	F	4	8	11
L	J, K	2	5	7

21

Question mappings

All 9 years

300

Linked examination marks

8

Exact matches

2 direct, 11 partial

2

Repeated numerical datasets

A-H and A-L

Key finding: This lecture is exceptionally exam-relevant. The A-H numerical dataset appears in 2017, 2019 and 2024, while the A-L dataset appears in 2016, 2018, 2021 and 2022. The 2023 problem uses a different dataset but the same method.

Report rule	Meaning
Exact	The lecture reproduces the question, dataset or requested definitions directly.
Direct	The lecture teaches the full method, but the examination uses a different dataset or broader wording.
Partial	Only part of the requested answer appears; another lecture or reference is still required.
Scope note	The report preserves the uploaded lecture content. Missing GERT, Gantt, float definitions and PERT pros/cons are identified rather than invented.

1. Lecture Coverage Map and Repetition Pattern

Slides	Core content	Question-bank use
1-2	PERT introduction; CPM fixed time versus PERT probability distribution	CPM-PERT distinction and PERT definition
3-4	Optimistic, most likely and pessimistic time; expected-time and variance formulas	Exact theory questions in 2017 and repeated short notes
5-8	Beta-to-normal approximation, central limit theorem, Z-score and normal table	Probability-of-completion questions
9-12	Ten-step PERT procedure, activity relations, scheduling, float, critical path and project variance	All numerical questions
13-21	Worked variance, project variance, standard deviation and deadline probability	Calculation and interpretation support
22	Example 9 - event-based PERT exercise	Additional practice; not an exact recurring exam dataset
23-24	Example 10 - A-H project	Exact repeat in 2017, 2019 and 2024
25-26	Example 11 - A-L project	Exact repeat in 2016, 2018, 2021 and 2022
27	Applications of PERT	Potential theory / short-note material
28	Reference only: advantages, shortcomings and CPM-PERT distinction	Not enough content for a complete answer

Question family	Years / frequency	Primary slides	Study implication
A-L PERT analysis	2016, 2018, 2021, 2022 (4 years)	25-26 plus 3-16	Highest priority exact repeat.
A-H PERT analysis	2017, 2019, 2024 (3 years)	23-24 plus 3-21	Highest priority exact repeat.
Other PERT numerical dataset	2023	3-16 and 22-26	Apply the same method to unfamiliar data.
Three-time estimates and variance	2017, 2018, 2019, 2024	3-4	Easy recurring theory marks.
Advantages / shortcomings	2017, 2018	28 only	Prepare separate notes; content is absent here.
GERT comparison	2016, 2021, 2022	2 and 28 only	Requires another source/lecture.
Float / slack terminology	2016, 2018-2020, 2023-2024	9-11 only	Use the CPM lecture for complete definitions.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024
Linked marks	54	35	35	35	10	28	33	35	35

2.1 Detailed Question-to-Slide Mapping: 2016-2018

Year	Original Sec./Q	Marks	QB page	Condensed question	Lecture slide(s)	Match	Coverage assessment
2016	A 2(a)	4	1	Define slack time and critical path.	9-12	Partial	The deck uses float and critical path in the PERT procedure, but does not give a complete standalone definition of slack.
2016	A 3(a)	10	2	Distinguish between CPM and PERT.	2, 28	Direct	Slide 2 gives the core distinction: CPM uses fixed activity times; PERT uses probability distributions. Slide 28 only points to a reference for a fuller comparison.
2016	A 3(b)	25	2	Analyze a 12-activity project by PERT; determine floats, critical path and 99% completion time.	3-16, 25-26	Exact	Slides 25-26 reproduce the same A-L dataset and task pattern; Slides 3-16 supply the formulas and probability procedure.
2016	B 8(a)	15	3	Define PERT and GERT; explain Gantt-chart objectives and preparation.	1-2	Partial	PERT and its probabilistic basis are present. GERT and Gantt-chart preparation are outside this deck.
2017	A 4(a)	8	5	Define optimistic time, most likely time, pessimistic time and variance.	3-4	Exact	All four requested definitions/formulas are stated together.
2017	A 4(b)	7	5	Write the advantages and shortcomings of PERT.	28	Partial	Slide 28 names this topic but does not contain the actual advantages or shortcomings.
2017	A 4(c)	20	5	Eight-activity PERT problem: network, critical path, schedule and 99% confidence.	3-21, 23-24	Exact	Slides 23-24 reproduce the same A-H dataset and tasks; Slides 3-21 provide the calculation method.
2018	B 8(a)	10	8	Short notes on CPM, free float, variance, optimistic time and pessimistic time.	2-4, 11	Partial	Variance and the two time estimates are covered exactly; CPM and free-float theory are not developed fully.
2018	B 8(b)	5	8	Advantages and shortcomings of PERT.	28	Partial	The deck refers students to a book rather than supplying the requested points.
2018	B 8(c)	20	8	Analyze a 12-activity PERT project; floats, critical path and 99% duration.	3-16, 25-26	Exact	The A-L dataset is reproduced directly on Slides 25-26.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 5.0 - PERT.pdf.

2.2 Detailed Question-to-Slide Mapping: 2019-2021

Year	Original Sec./Q	Marks	QB page	Condensed question	Lecture slide(s)	Match	Coverage assessment
2019	A 4(a)	10	9	Define free float, independent float, dummy activity, optimistic time and pessimistic time.	3, 9-11	Partial	Optimistic and pessimistic time are exact; the three network/float terms are only referenced, not fully defined.
2019	A 4(b)	25	10	Eight-activity PERT problem with schedule, total/free/independent float and 99% start time.	3-21, 23-24	Exact	Slides 23-24 contain the same A-H dataset and nearly identical requirements.
2020	A 2(a)	10	12	Define critical float, slack time, PERT, free float and independent float.	1-2, 9-11	Partial	PERT is covered and float is used in the procedure, but the float definitions are not complete in this deck.
2021	A 4(c)	20	16	Analyze the A-L project using PERT; total float, critical path and 99% completion.	3-16, 25-26	Exact	The same A-L dataset and task sequence appear on Slides 25-26.
2021	B 8(b)	8	17	Define GERT and LSM; distinguish GERT from CPM or PERT.	2, 28	Partial	The PERT side of the comparison is present; GERT and LSM are not taught here.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 5.0 - PERT.pdf.

2.3 Detailed Question-to-Slide Mapping: 2022-2024

Year	Original Sec./Q	Marks	QB page	Condensed question	Lecture slide(s)	Match	Coverage assessment
2022	A 4(a)	25	18	Analyze the A-L project by PERT and find 99% completion time.	3-16, 25-26	Exact	The question repeats the A-L dataset shown on Slides 25-26.
2022	B 8(b)	8	19	What is GERT? Differentiate between GERT and PERT.	2, 28	Partial	PERT variability is described, but the deck has no GERT definition.
2023	A 4(a)	25	20-21	Analyze a different A-K project using PERT; floats, critical path and 99% duration.	3-16, 22-26	Direct	The complete method and comparable worked examples are present, but the 2023 A-K dataset is not reproduced.
2023	A 4(b)	10	21	Define total float, free float, independent float, early start and late start.	9-11	Partial	The PERT steps mention ES, LS and float, but do not give full exam-ready definitions for all five terms.
2024	A 4(a)	10	23	Define critical path, free float, independent float, variance and slack time.	4, 9-16	Partial	Variance and critical-path use are covered; complete definitions of all float/slack terms are missing.
2024	A 4(b)	25	24	Eight-activity PERT problem: network, critical path, activity schedule and 99% start time.	3-21, 23-24	Exact	The A-H dataset and task wording closely match Slides 23-24.

QB = original question-bank PDF page. Slide numbers refer to the uploaded 5.0 - PERT.pdf.

3. Exact Dataset Repeats and Visual Verification

3.1 A-H dataset: 2017, 2019 and 2024

The activity table on Lecture Slides 23-24 matches the repeated A-H examination dataset, including B pessimistic time 26 and the 30-week / 99% requirement.

Example: 10 (ref. V.K. Kapoor, pg 16.49)

- A small project consisting of eight activities has the following characteristics:

Time estimates (in weeks)

Activity	Preceding activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	4	12
B	-	10	12	26
C	A	8	9	10
D	A	10	15	20
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Lecture Slide 23

Question-bank confirmation

The 2024 question reproduces the same predecessors and three time estimates. The same dataset also appears in 2017 and 2019.

The screenshot shows a question bank page with a table of activity time estimates and a list of questions. The table is titled 'Time estimates (in weeks)' and has columns for Activity, Preceding activity, Optimistic time, Most likely time, and Pessimistic time. The questions are numbered 1 to 15 and cover topics such as project network, activity schedule, project completion time, resource loading, and subcontractors.

Activity	Preceding activity	Optimistic time	Most likely time	Pessimistic time
A	-	2	4	12
B	-	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	B, C	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

Questions:

- Draw the PERT network for the project and determine the critical path for the project.
- Prepare the activity schedule for the project.
- If the project manager wants to be 99% sure that the project is completed within 30 weeks, how many weeks before that date should the project work be started?
- Describe how a resource loading chart helps project managers identify potential overallocation or underutilization of resources.
- What are the suitability of cost reimbursement contracts? Briefly explain the main aspects of each type contract.
- Consider a project with 8 activities as shown in the Table. The predecessors and durations of each activity are given along with the number of labour (considered as resources) required for each activity. The organisation has only 10 resources available. Allocate and level the resources to achieve maximum benefit, ensuring that the critical path does not increase (including floatings).
- Describe the 'pay-if-paid' clause and explain why courts generally reject this clause with default.
- Define attendance. Identify and explain the different types of subcontractors involved in a construction project and the specific roles they play in the overall construction process.
- A utility contractor must install a 1200 m water main along a road. Work is repetitive and in sequence.
 - Excavation
 - Bedding and sand
 - Pipe laying
 - Backfilling
 - Surface reinstatement

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Question bank 2024, PDF page 24

3.2 A-L dataset: 2016, 2018, 2021 and 2022

Lecture Slides 25-26 reproduce the A-L dataset and ask for network, total float, critical path and 99% completion duration.

Example: 11 (ref. V.K. Kapoor, pg 16.51)

- The information is given (time estimates are in days):

Activity	Preceding activity	Optimistic (a)	Most Likely (m)	Pessimistic (b)
A	-	2	2	2
B	-	1	3	7
C	A	4	7	8
D	A	3	5	7
E	B	2	6	9
F	B	5	9	11
G	C, D	3	6	8
H	E	2	6	9
I	C, D	3	5	8
J	G, H	1	3	4
K	F	4	8	11
L	J, K	2	5	7

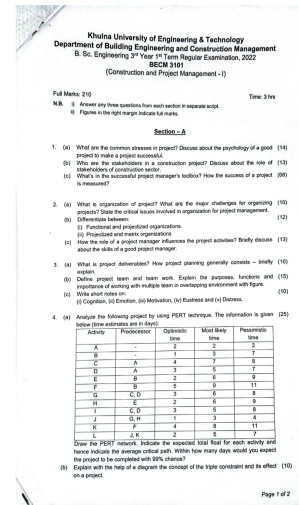
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3-25

Lecture Slide 25

Question-bank confirmation

The 2022 question is an exact repeat. The same dataset is visible in the 2016, 2018 and 2021 papers.



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Question bank 2022, PDF page 18

4. Formula Sheet and Verified Numerical Results

Purpose	Formula / rule
Expected activity time	$t_e = (a + 4m + b) / 6$
Activity variance	$\text{variance} = [(b - a) / 6]^2$
Project expected duration	Sum expected times along the critical path
Project variance	Sum activity variances along the critical path
Project standard deviation	$\text{sigma} = \text{square root of project variance}$
Deadline probability	$Z = (\text{due date} - \text{expected completion date}) / \text{project standard deviation}$
99% confidence duration	Expected duration + 2.33 x project standard deviation

Repeated dataset	Verified result derived from the uploaded formulas
A-H (Slides 23-24)	Expected activity durations: A=5, B=14, C=9, D=15, E=8, F=9, G=4, H=5 weeks. Critical path = A-D-G-H. Expected project duration = 29 weeks. Critical-path variance = 6; standard deviation = 2.449 weeks. 99% duration = about 34.7 weeks, so for a 30-week target the project should start about 4.7 weeks earlier.
A-L (Slides 25-26)	Critical path = B-F-K-L. Expected project duration = 24.67 days. Critical-path variance = 4.056; standard deviation = 2.014 days. 99% completion duration = about 29.36 days.

Variability in Activity Times

Estimate follows beta distribution

Expected time:

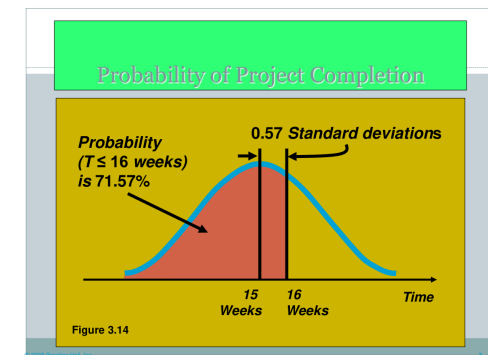
$$t_e = (a + 4m + b) / 6$$

Variance of times:

$$\sigma^2 = [(b - a) / 6]^2$$

(iii) Probability computation using Z-distribution table:

- Calculate $Z = (t - t_e) / \sigma$
- From Z-distribution table, the value corresponding to Z gives the required probability.



5. Exam-Focused Study Plan, Gaps and Corrections

Priority	What to study	Slides	Why
1	Master three-time estimates, expected time and variance	3-4	Direct theory marks and the basis of every calculation.
2	Learn probability workflow: project variance, standard deviation and Z-table	5-8, 12-21	Required for every 99% completion question.
3	Solve the A-H dataset without looking at the answer	23-24	Exact repeat in three years.
4	Solve the A-L dataset without looking at the answer	25-26	Exact repeat in four years.
5	Practise one unfamiliar dataset	22 and 2023 QB pages 20-21	Prepares for changed numbers and networks.
6	Prepare separate one-page notes on GERT, Gantt, float types and PERT pros/cons	Other lectures/reference	These are repeatedly asked but incomplete in this deck.

Source issue / limitation	Recommended treatment
Slide 1 writes "Proframme"	The report uses the standard title "Programme Evaluation and Review Technique" without altering the source slide.
Slide 11 mixes tm and te in the scheduling steps	Use expected time te consistently for PERT forward/backward calculations. Treat the slide wording as an internal inconsistency.
Slide 28 lists advantages, shortcomings and distinction only as a reading reference	Do not rely on Slide 28 alone for those exam questions; obtain the referenced material or use the CPM/PERT comparison lecture.
GERT and Gantt questions	Not supported by this deck; they are deliberately marked Partial.
Float/slack definitions	The deck uses these ideas but does not provide a full definition set. Review the CPM lecture.

Recommended answer order for a numerical PERT problem: draw the network -> calculate te and variance for every activity -> forward pass -> backward pass -> floats -> critical path -> project variance and standard deviation -> Z/probability or 99% completion duration.

Applications of PERT

- Successfully used in construction industry, particularly in building of large structures.
- Scheduling of govt. and commercial aircrafts, installing fixed assets & plant lay-out.
- Also been used in general administration of firm, streamlining paper work, making major programme planning and corporate profit plan development.
- Controlling the developments and launching of new products.

See (ref. V.K. Kapoor, pg 16.44 to 16.46)

- Advantages of PERT
- Shortcomings of PERT
- Distinction between PERT & CPM